

SWE 4739

Embedded Software Development

Lecture 6

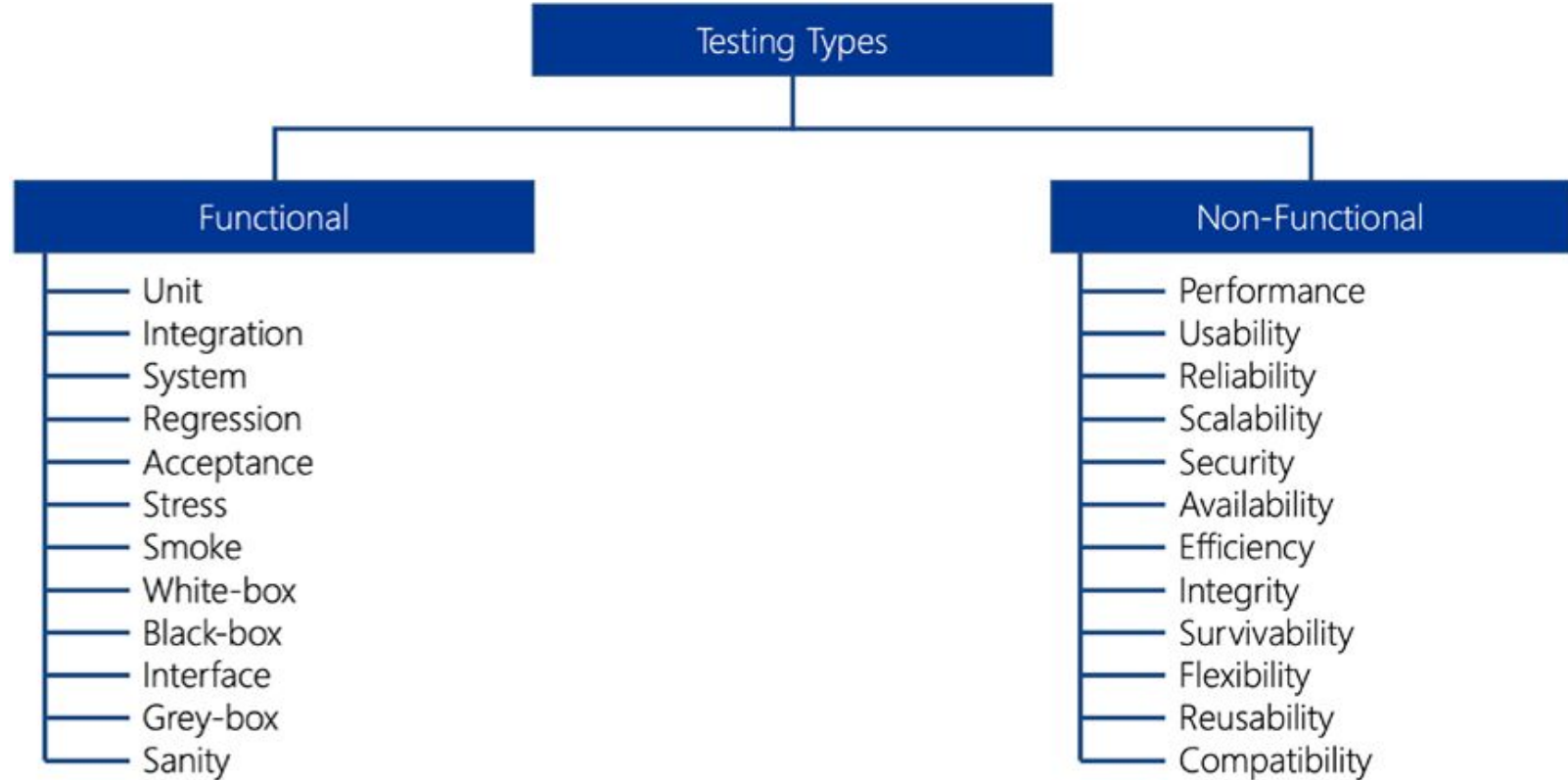
Testing, Verification, and TDD

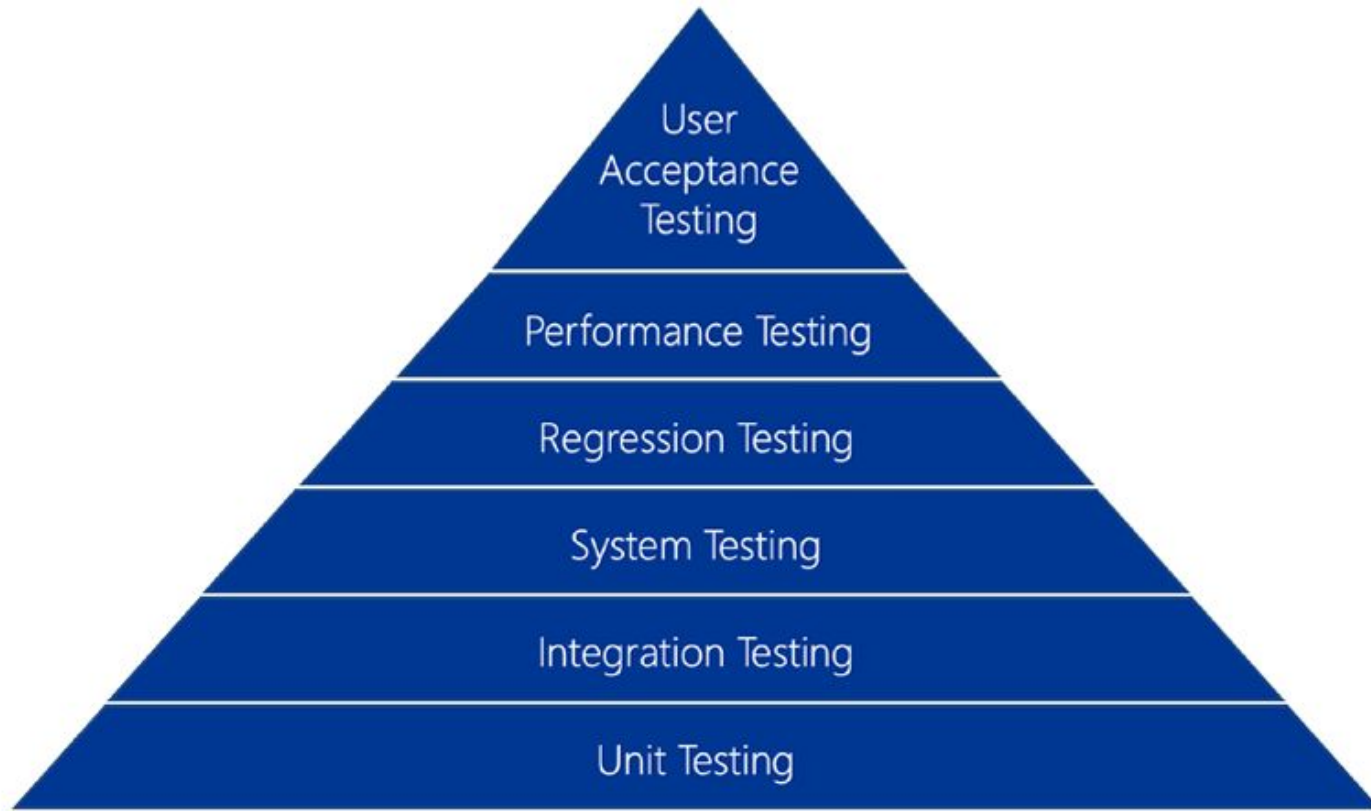
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Testing (Example)

```
def calculate_total(cart_items):
    total = 0
    for price, quantity in cart_items:
        total += price * quantity
    return total

def apply_discount(total, discount_code):
    if discount_code == "SAVE10":
        return total * 0.9
    elif discount_code == "SAVE20":
        return total * 0.8
    return total

def apply_tax(amount):
    return amount * 1.05    # 5% tax

def checkout(cart_items, discount_code):
    total = calculate_total(cart_items)
    discounted = apply_discount(total, discount_code)
    final_amount = apply_tax(discounted)
    return final_amount
```

Unit Testing

```
def test_apply_discount():  
    assert apply_discount(100, "SAVE10") == 90  
    assert apply_discount(100, "SAVE20") == 80  
    assert apply_discount(100, "INVALID") == 100
```

Integration Testing

```
def test_checkout():  
    cart = [(100, 1), (50, 2)] # total = 200  
    result = checkout(cart, "SAVE10")  
    # 200 -> 180 -> 189  
    assert result == 189
```

Regression Testing

Regression testing is the process of re-running previously executed tests after making changes to the system to ensure that existing functionality still works correctly.

After adding a new discount feature, previously working discount calculations must still produce correct results.

Performance Testing

Performance testing evaluates how efficiently a system operates under specific conditions, focusing on speed, responsiveness, and scalability.

Example:

Testing whether a checkout system can process thousands of items within an acceptable time limit.

Reference:

1. Embedded Software Design: A Practical Approach to Architecture, Processes, and Coding Technique – Jacob Beningo

Thank You!